

Lecture 5 Amorphous and Semi-Crystalline Materials

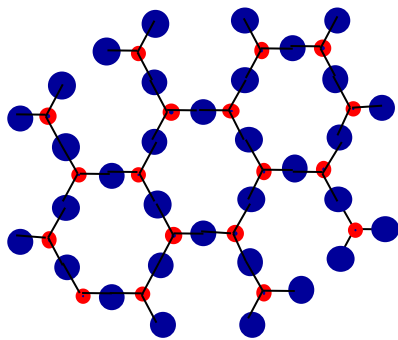
Upcoming Dates:

- HW3 – Due today
- HW4 – Module 3 Prelab – Due Thursday
- Module 2 Expert Worksheet – Due Tuesday.
 - Upload to Canvas before class
 - Bring a copy to class to use for our activity
- Module 3 – Labs starts Next week on Monday

Review: Materials and Packing

Crystalline materials...

- atoms pack in periodic, 3D arrays
- typical of:
 - metals
 - many ceramics
 - some polymers



crystalline SiO₂

Adapted from Fig. 3.40(a),
Callister & Rethwisch 3e.

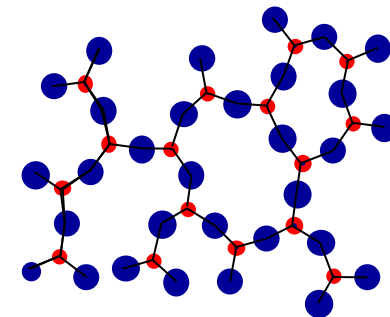
• Si

• Oxygen

Noncrystalline materials...

- atoms have no periodic packing
- occurs for:
 - complex structures
 - rapid cooling

"Amorphous" = Noncrystalline



noncrystalline SiO₂

Adapted from Fig. 3.40(b),
Callister & Rethwisch 3e.

Polymers - Historical

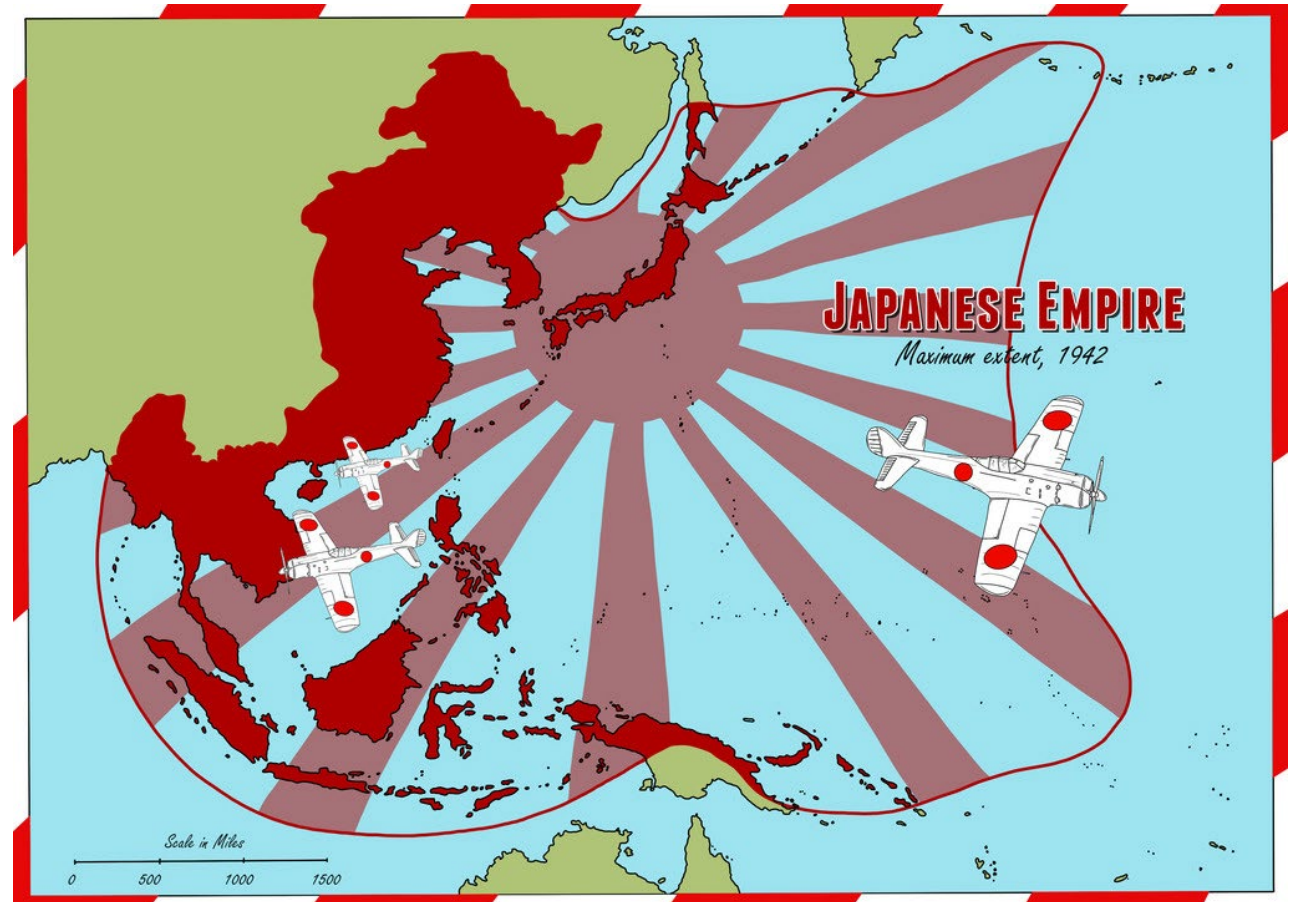


The invention of the automobile in the late 19th century, led to a boom for rubber plantations



The plantations of the enormous **Southeast Asian rubber** industry were so efficient that the **1930s** enjoyed a global glut of **rubber exports**

SOUTHEAST ASIA



- **June 1940, rubber had become so crucial that President Roosevelt declared it a strategic material.**
- **In 1941 Japan seized the rubber lands of Southeast Asia,**
- **U.S. production of synthetic rubber increased a hundredfold—from 9,000 short tons in 1941 to 919,000 tons in 1945, at which point synthetic rubber met four-fifths of America's needs.**

<http://www.answers.com/topic/rubber>

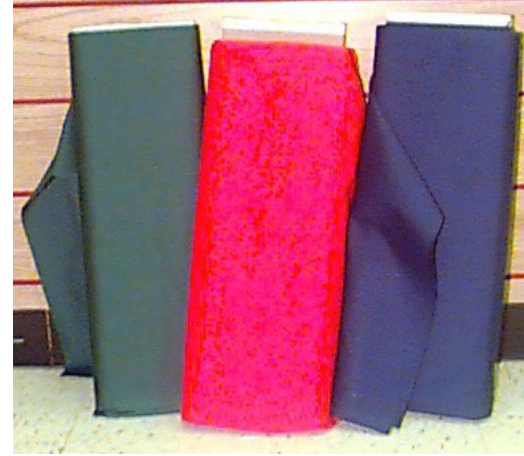
Natural Polymers

- skin, muscle, blood vessels, nerves, starches, proteins
- silk, cotton, wool...
- natural rubber

Synthetic Polymers - Everywhere



Polyethylene



Polyester



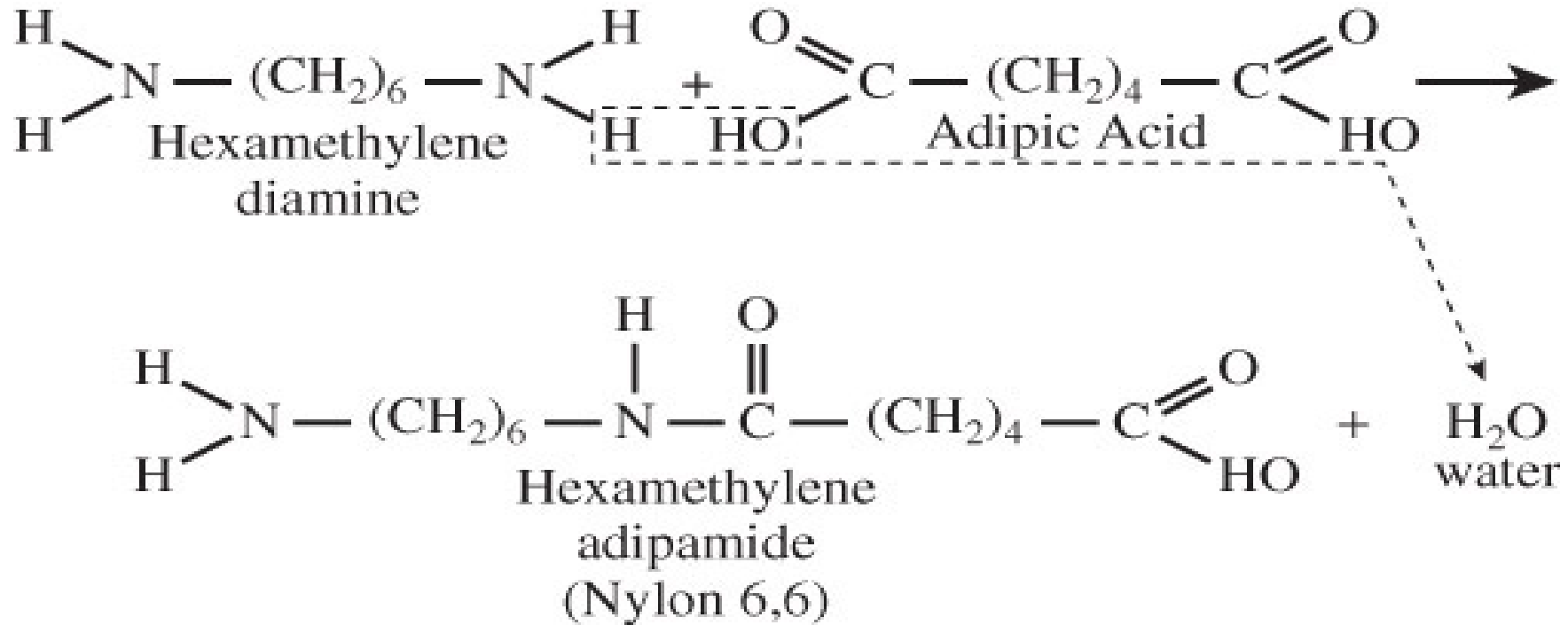
Nylon and Polyethylene



Head light cover--- Polycarbonate

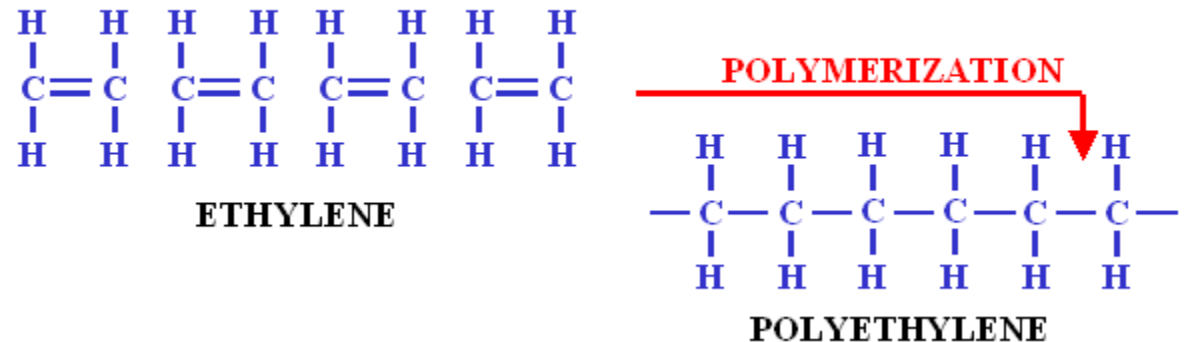
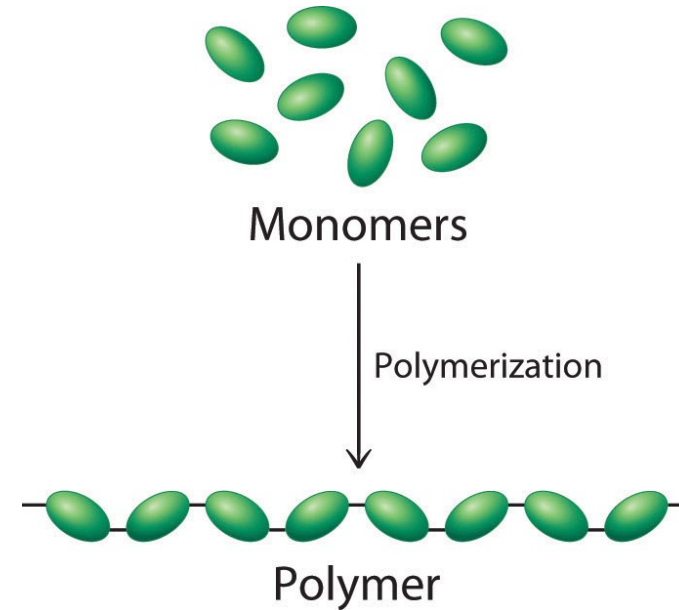
Polymerization Reactions

Example



Polymer Macromolecules

- poly: many;
- mer: unit, part



- Molecular Weight of a Polymer (MW)
 - represents a statistical mean in a distribution of polymer chains of different lengths about that mean.
 - not exactly the same for one polymer
- Degree of polymerization (DP)
 - can be very different for even one polymer

$$DP = \frac{\text{Average Molecular Weight}}{\text{Monomer Molecular Weight}} = \frac{MW_{ave}}{MW_{monomer}}$$

example: UHMWPE

PE: How Does Molecular Weight of a Polymer Affect Performance?

Low Density Polyethylene
(LDPE)



High Density Polyethylene
(HDPE)



Ultra High Molecular Weight Polyethylene
(UHMWPE)

Total hip replacement

1 Damaged head of femur is removed



2 Ball prosthesis is cemented into hollowed-out femur



3 Socket prosthesis is prepared and connected to ball prosthesis



Coffey infographic with information from the Arthritis Foundation

Hip Arthroplasty History

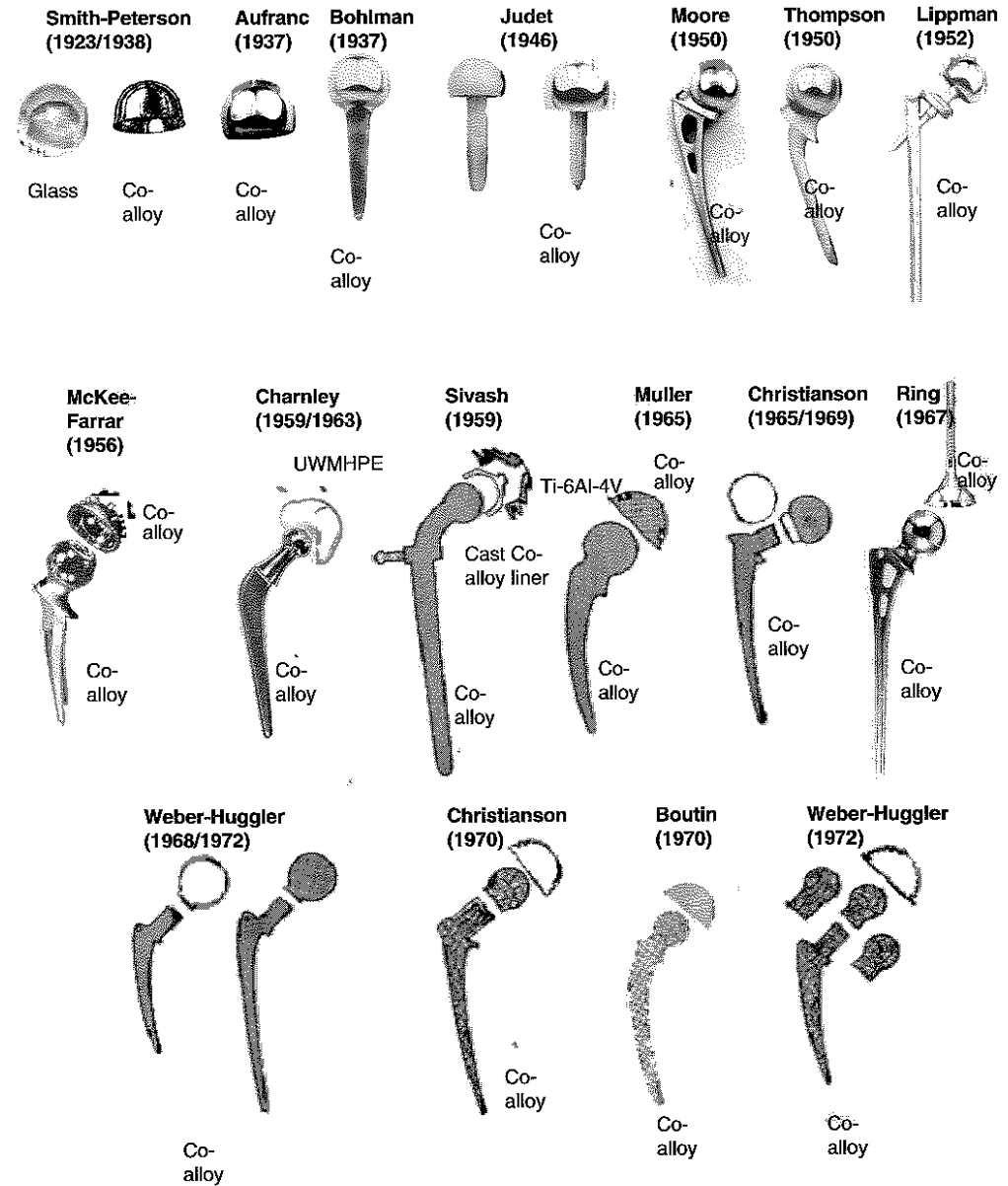
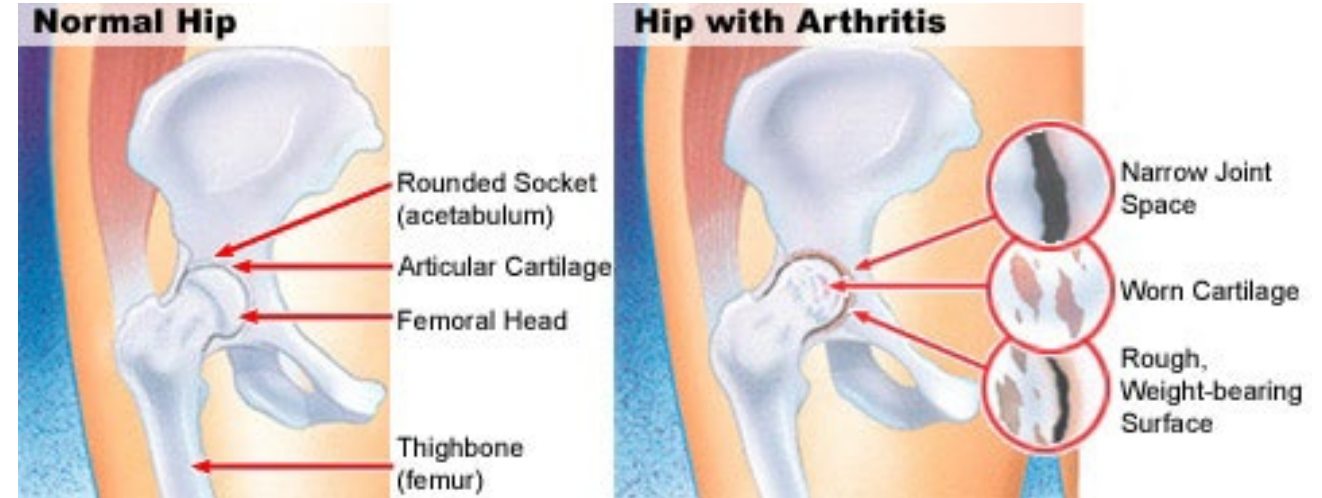


FIG. 2. The history of total hip arthroplasty is particularly pertinent to biomaterials science because it is one of the best illustrations of how an implant first used over a century ago has evolved to the highly successful status it has, primarily because of advances in biomaterials.



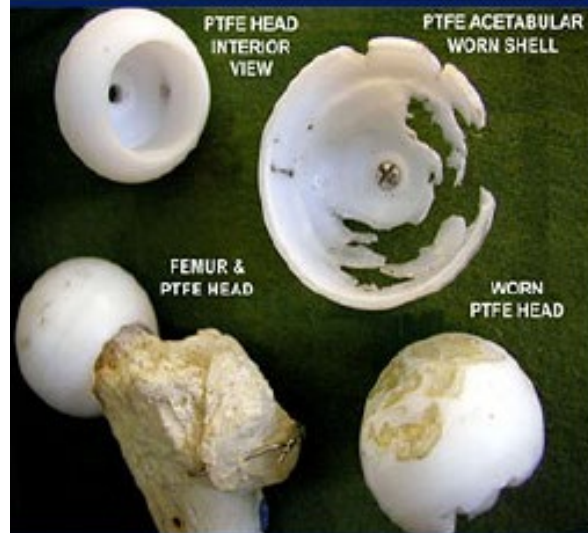
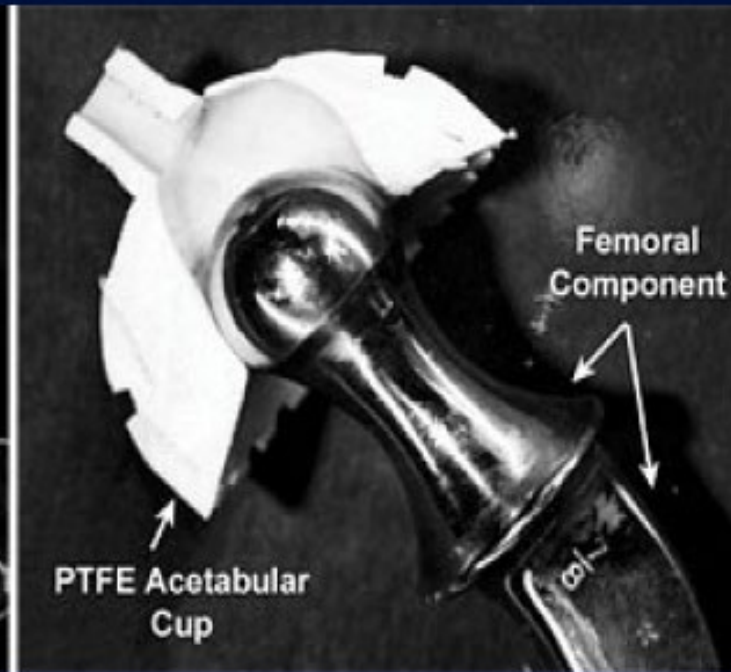
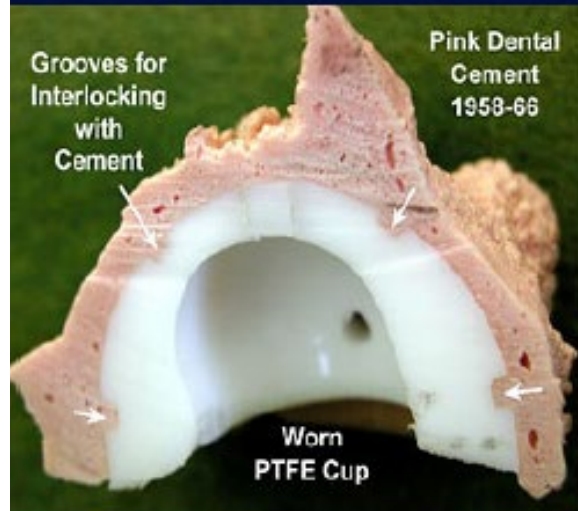
General view of Thompson and Austin Moore prosthesis
TH: Thompson Prosthesis, AM: Austin Moore prosthesis

Osteoarthritis is the most common reason for hip replacement



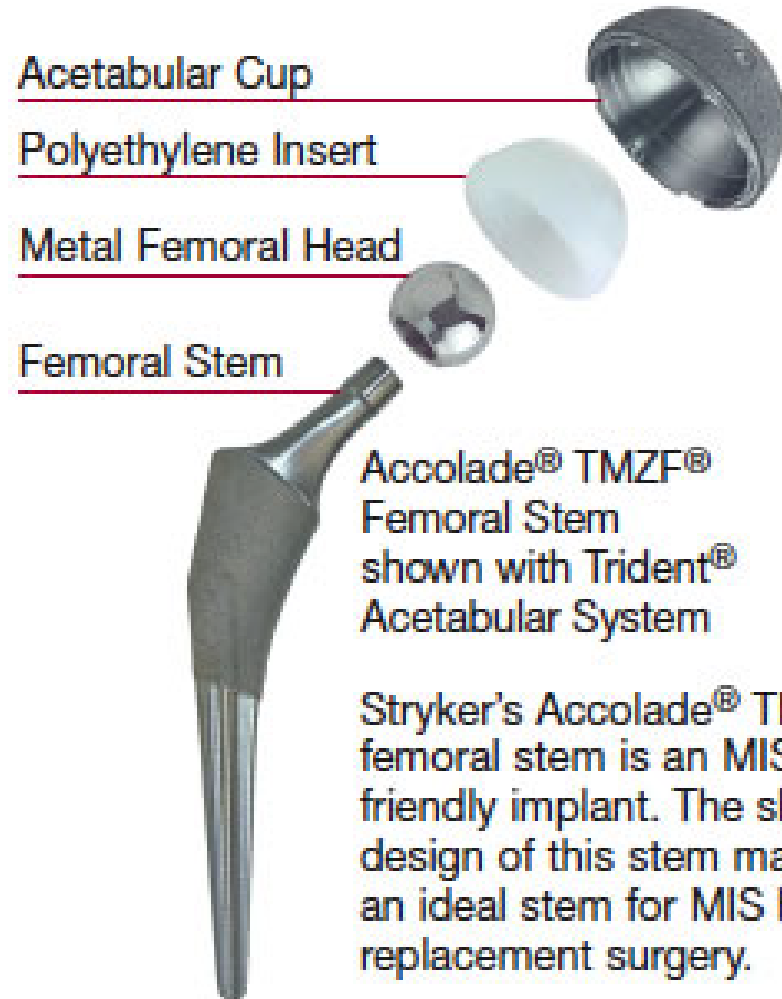
- Initially, only the femoral head was replaced using cementless (Moore) and cemented (Thompson) prosthesis.
- These devices, however, could only be used when the acetabulum was unchanged, limiting the applications to femoral neck fracture.

Wear of a PTFE acetabular cup



In the mid-1950s, Dr. John Charnley, MD, used PTFE for the hip cup, but it wore extensively, and by 1962 he had to retrieve all of his implants. He subsequently replaced the PTFE with UHMWPE, which exhibited reduced wear and increased joint lifetime. (C. Migliaresi slide)

Current: Total Hip Anthroplasty

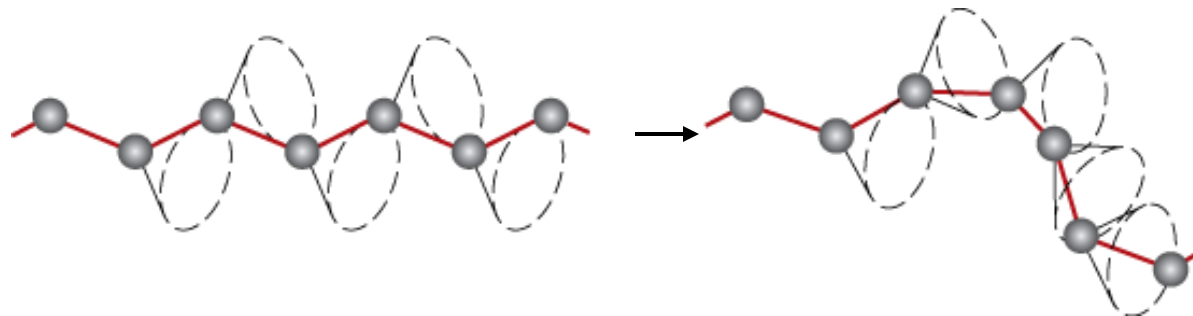


Stryker's Accolade® TMZF® femoral stem is an MIS friendly implant. The slim design of this stem makes it an ideal stem for MIS hip replacement surgery.

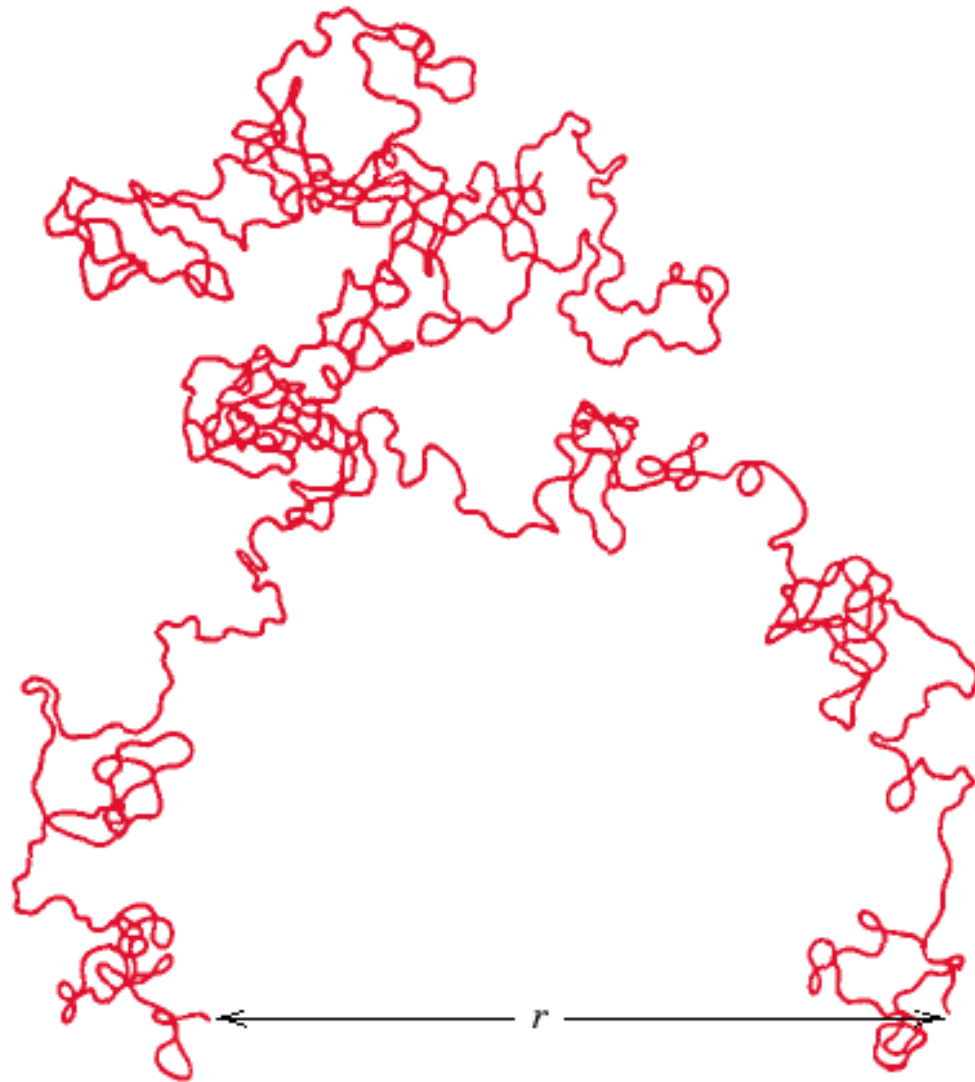
Polymers – Molecular Shape

Molecular Shape (or **Conformation**)

- chain bending and twisting are possible by rotation of carbon atoms around their chain bonds
- As a result, it's not necessary to break chain bonds to alter molecular shape



Adapted from Fig.
4.5, Callister &
Rethwisch 3e.



Adapted from Fig.
4.6, *Callister &
Rethwisch 3e.*

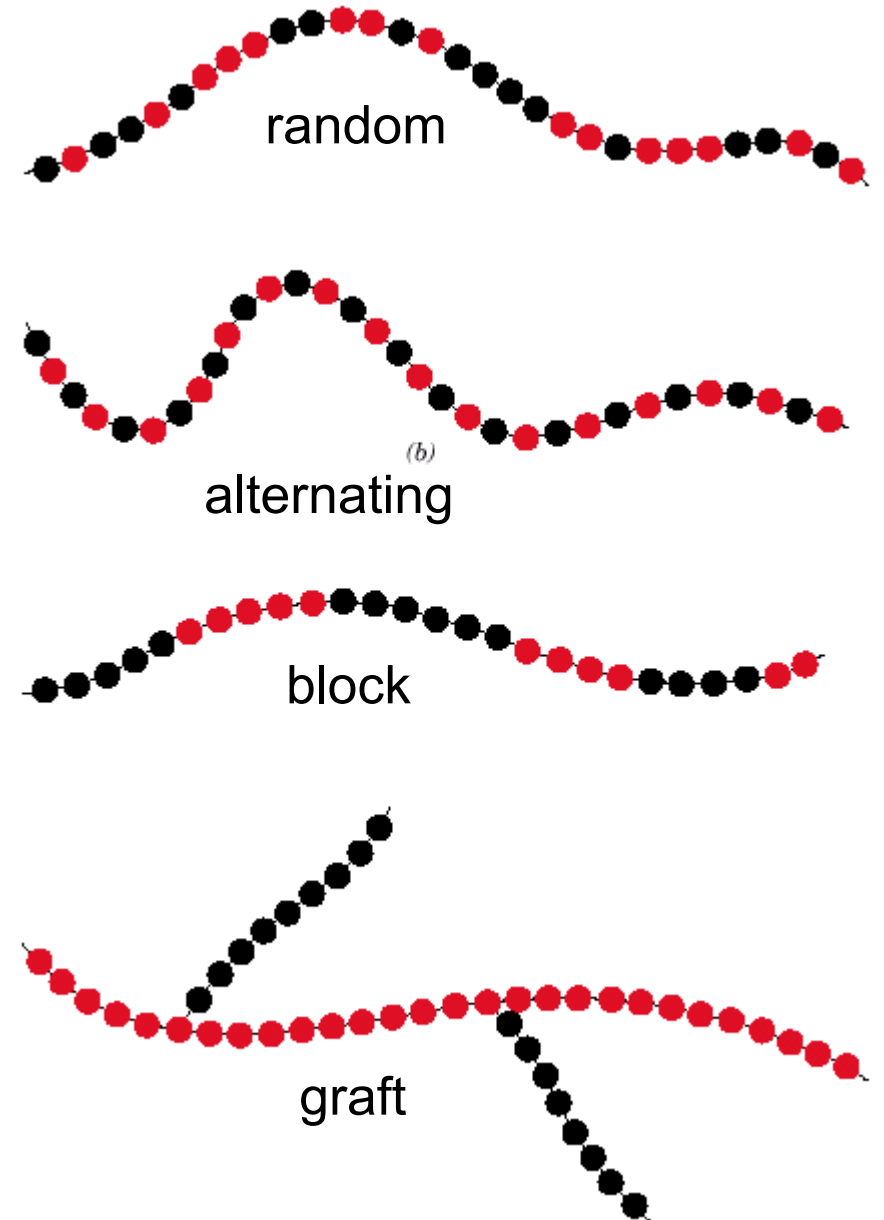
Copolymers

Def: two or more monomers polymerized together

- **random** – A and B randomly positioned along chain
- **alternating** – A and B alternate in polymer chain
- **block** – large blocks of A units alternate with large blocks of B units
- **graft** – chains of B units grafted onto A backbone

A – ●

B – ●



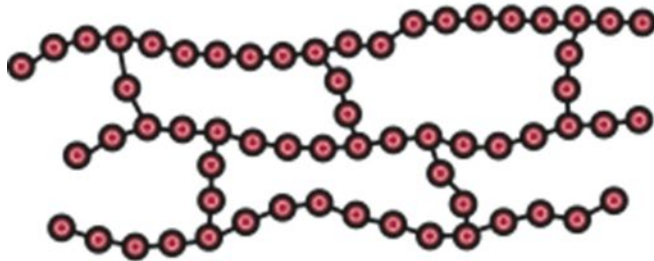
Polymer Structure



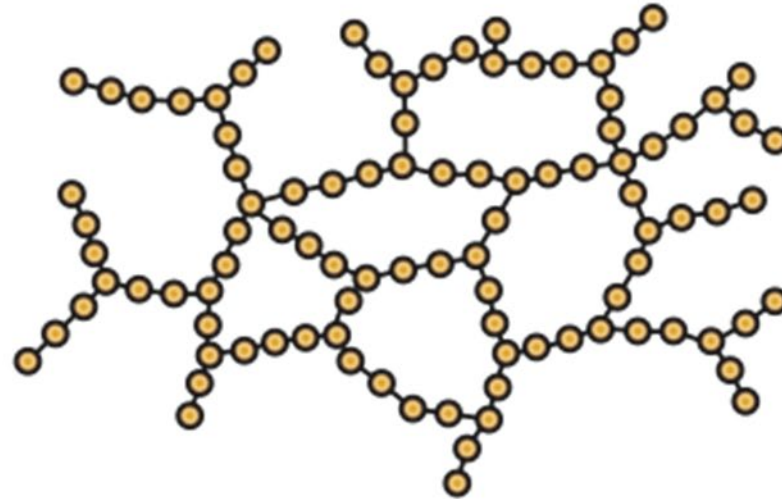
(a) Linear



(b) Branched

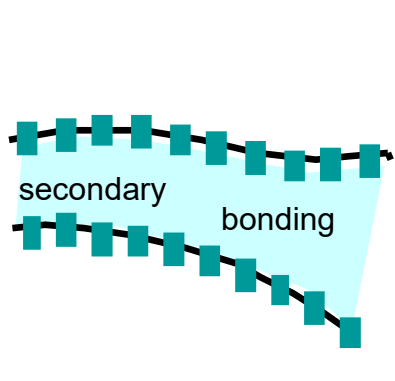


(c) Cross-linked



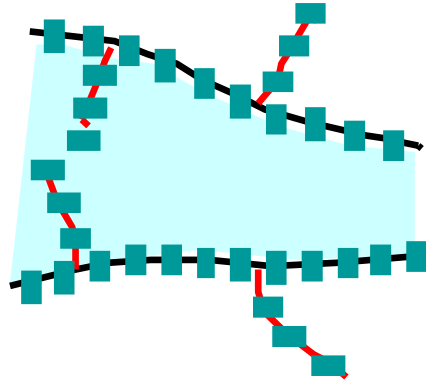
(d) Network

Molecular Structures for Polymers



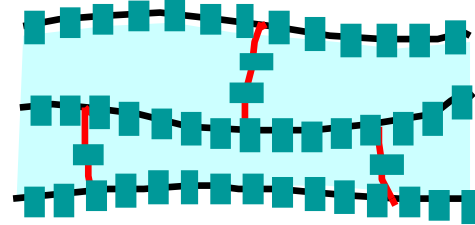
Linear

HDPE, PVC, Nylon



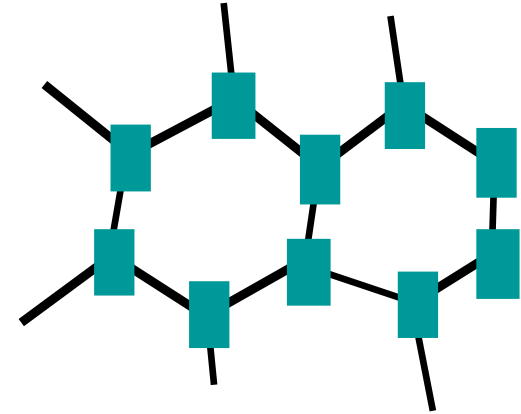
Branched

LDPE



Cross-Linked

vulcanized rubber



Network

epoxies

Polymer Classifications

1. Thermoplastic

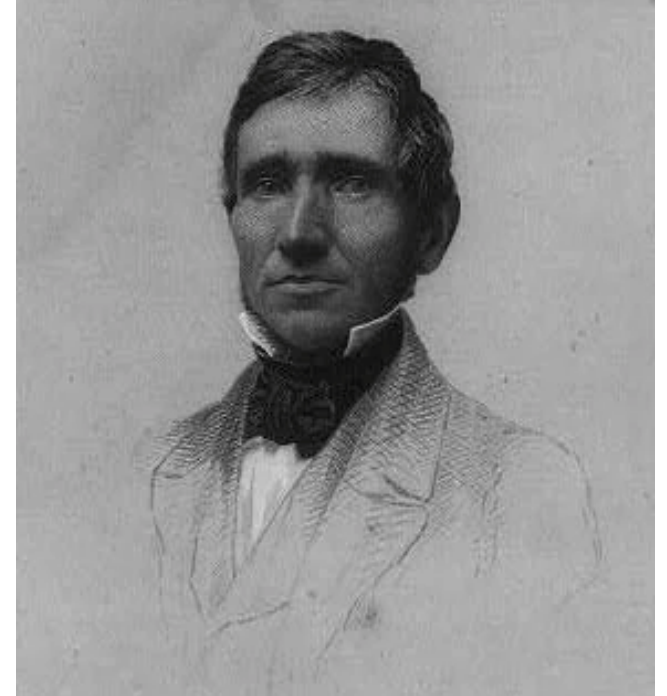
- Soften when heated, harden when cooled
- Most linear and branched polymers
- Can be recycled
- More \$\$ to make
- HDPE, PVC, ABS

2. Thermosets

- Permanently hard during their formation
- Don't soften upon heating
- Most cross-linked and all network polymers
- Can't be recycled
- resins, some rubbers

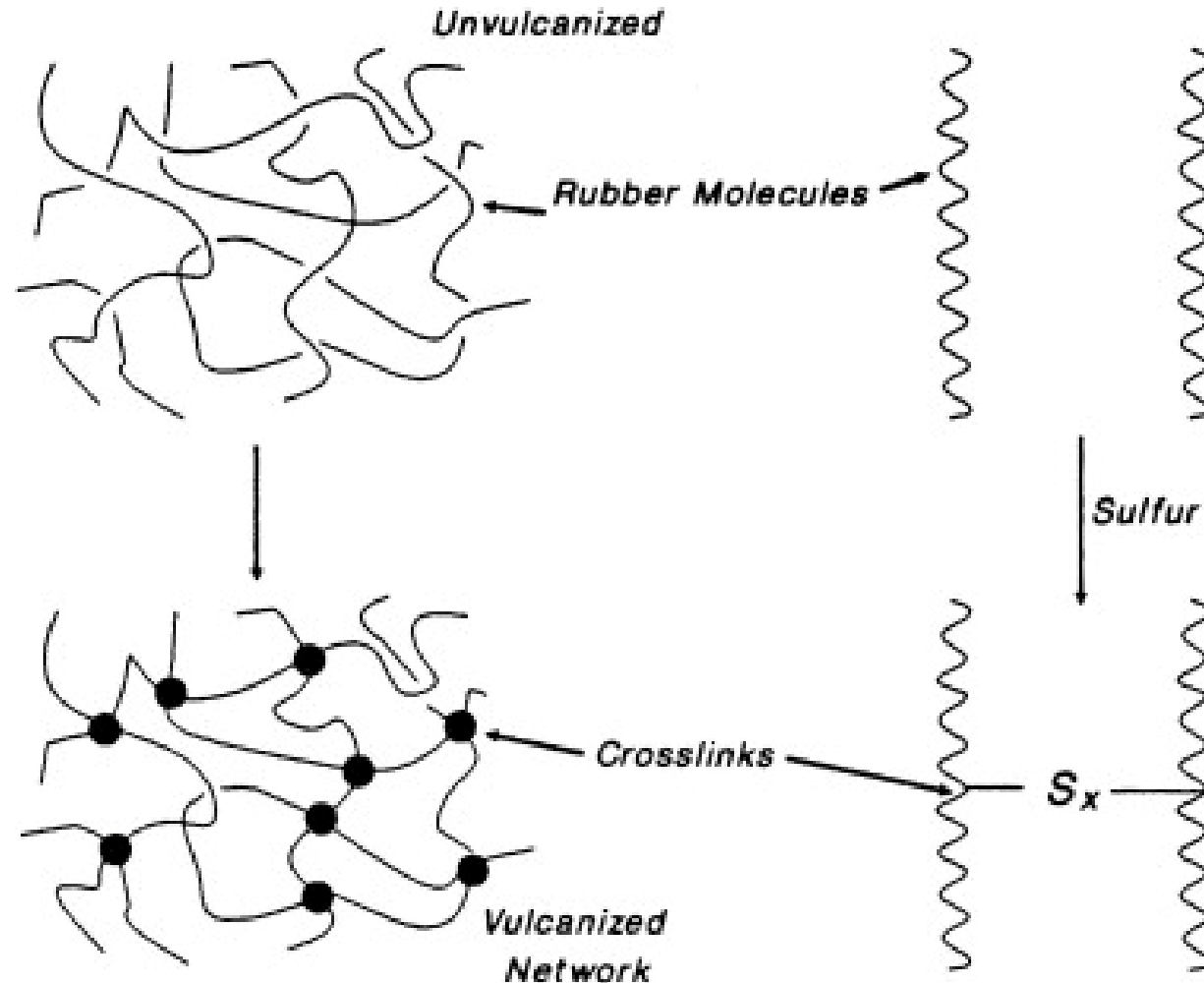
Charles Goodyear

- 1800 - 1860
- American inventor of the vulcanization process that made possible the commercial use of [rubber](#).
- In 1839 he accidentally dropped some India rubber mixed with sulfur on a hot stove and so discovered [vulcanization](#).
- Although his invention made millions for others, at his death he left debts of some \$200,000.
- Goodyear was founded in 1898 by Frank Seiberling



Unvulcanized rubber is generally not strong, does not retract essentially to its original shape after a large deformation, and it can be very sticky.

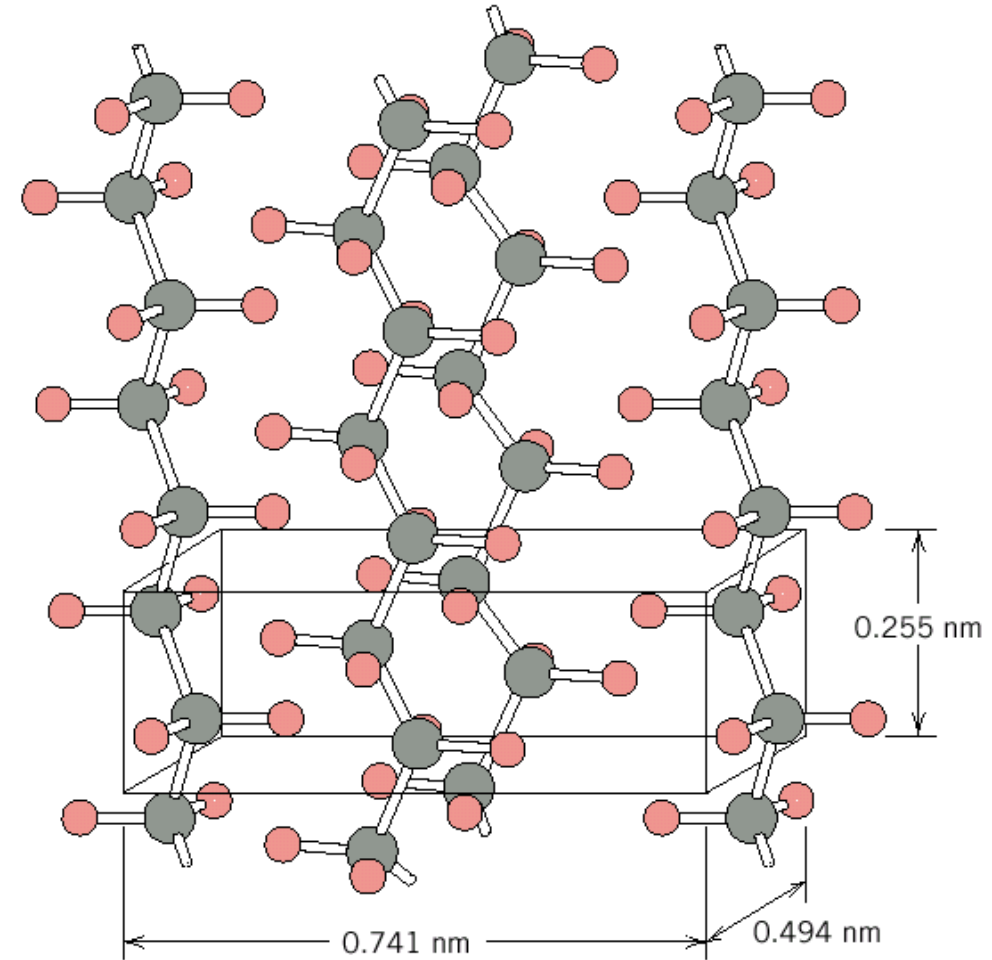
Thermoplastic:



Thermoset:

Crystallinity in Polymers

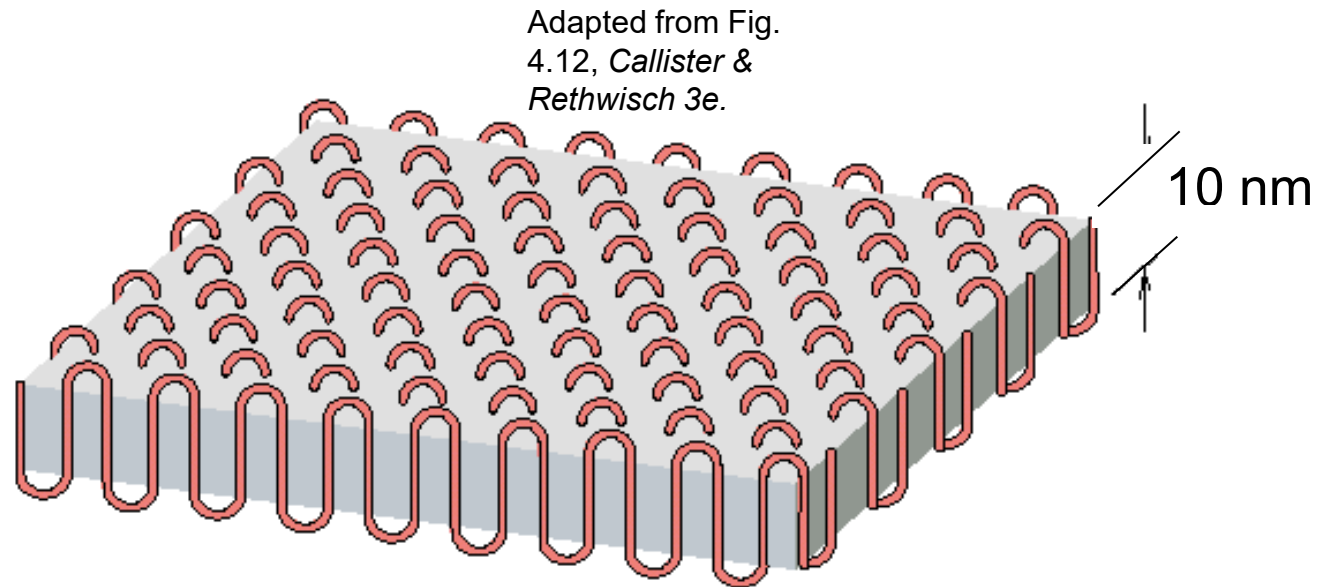
- Ordered atomic arrangements involving molecular chains
- Crystal structures in terms of unit cells
- Example shown
 - polyethylene unit cell



Adapted from Fig.
4.10, *Callister &
Rethwisch 3e.*

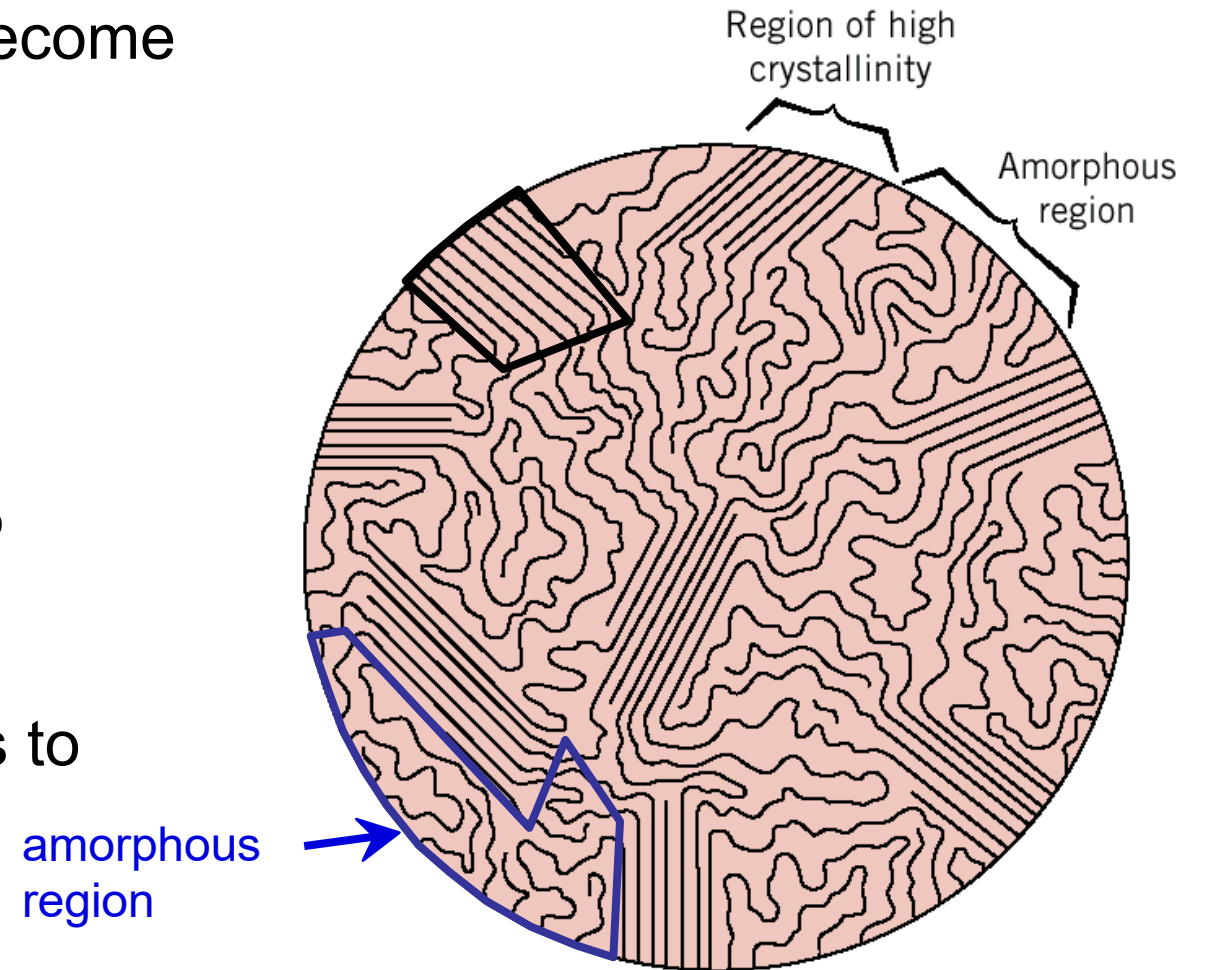
Polymer Crystallinity

- Crystalline regions
 - thin platelets with chain folds at faces
 - Chain folded structure



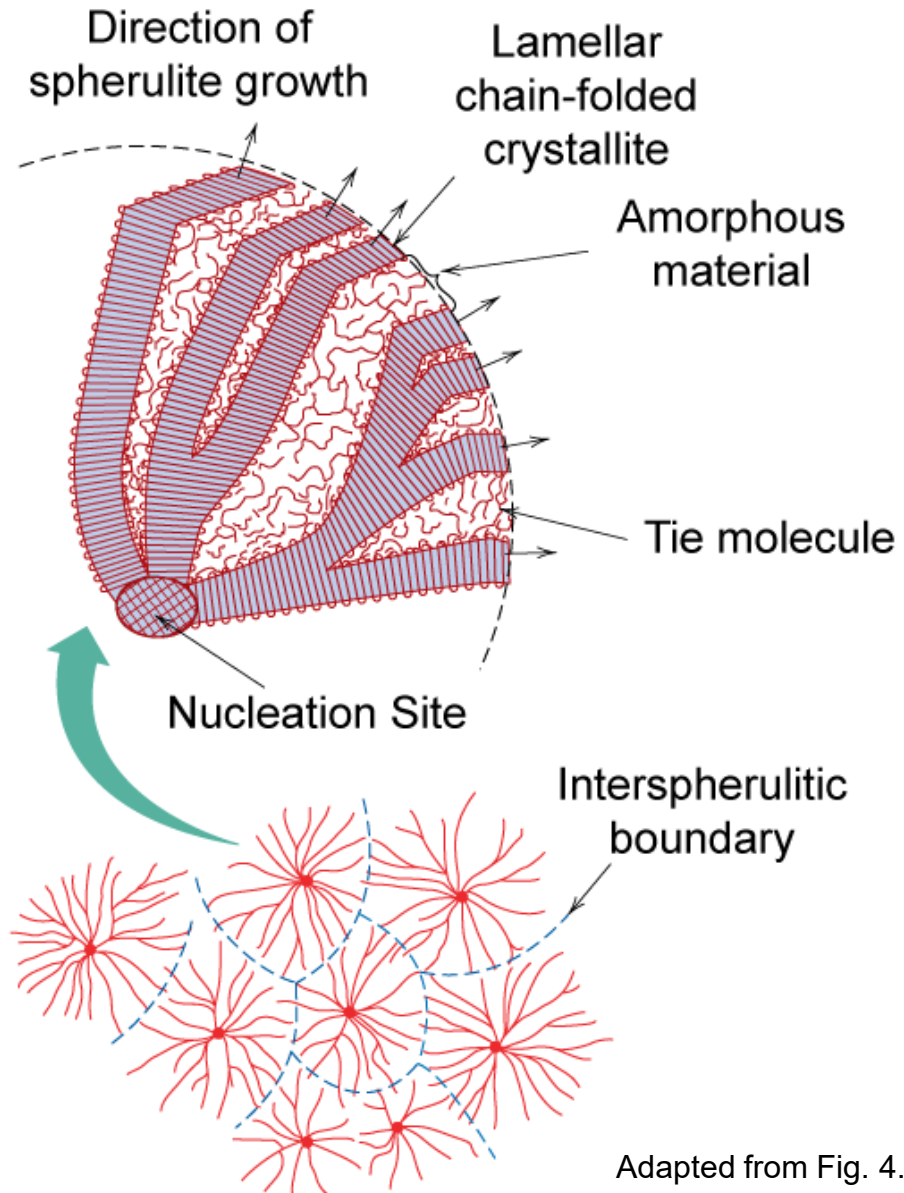
Polymers are rarely 100% crystalline

- Difficult for all regions of all chains to become aligned
- Degree of crystallinity expressed as %crystallinity.
- Some physical properties depend on % crystallinity.
- Heat treating causes crystalline regions to grow and % crystallinity to increase



Adapted from Fig. 14.11, *Callister 6e*. (Fig. 14.11 is from H.W. Hayden, W.G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behavior*, John Wiley and Sons, Inc., 1965.)

Semicrystalline Polymers



- Some semicrystalline polymers form **spherulite** structures
- Alternating chain-folder crystallites and amorphous regions
- Spherulite structure for relatively rapid growth rates

Which polymer in each of the following pair is more likely to crystallize?

- a. Linear polypropylene vs. cross-linked polyisoprene
- b. Block copolymer poly(acrylonitrile-isoprene) vs. graft copolymer poly(chloroprene-isobutylene).

MIT Engineers Create the “Impossible” New Material Stronger Than Steel and As Light as Plastic

<https://scitechdaily.com/mit-engineers-create-the-impossible-new-material-that-is-stronger-than-steel-and-as-light-as-plastic/>



MIT Engineers Create the “Impossible” New Material Stronger Than Steel and As Light as Plastic

- Traditionally, polymer chains grow by adding new molecules onto their ends.
 - Once formed, polymers can be shaped into three-dimensional objects.
- A new polymerization process:
 - Generate a two-dimensional sheet called a polyaramide. Building blocks contains a ring of carbon and nitrogen atoms.
 - These monomers can grow in **two dimensions**, forming disks.
 - These disks held together by hydrogen bonds between the layers.

Polo Ralph Lauren

Team USA 2022 Winter Olympics Opening Ceremony Anorak

\$1,998.00



skyscraper



- Insulation:
 - 80% polyester
 - 20% recycled polyester
 - Patented underneath Intelligent Insulation by Skyscrape.

<https://www.youtube.com/watch?v=C8wxdzMFeNo>

Skyscraper Intelligent Insulation

- An innovative patented fabric that responds to changes in temperature without the use of battery-powered or 'wired' technology.
- As the temperature drops, the Skyscraper Intelligent Insulation expands, offering more insulation and warmth.

Unfortunately, we are identifying some negative results from the “plastics age”





Dumped into the ocean in 1976 and still, this piece of plastic hasn't degraded at all. :(

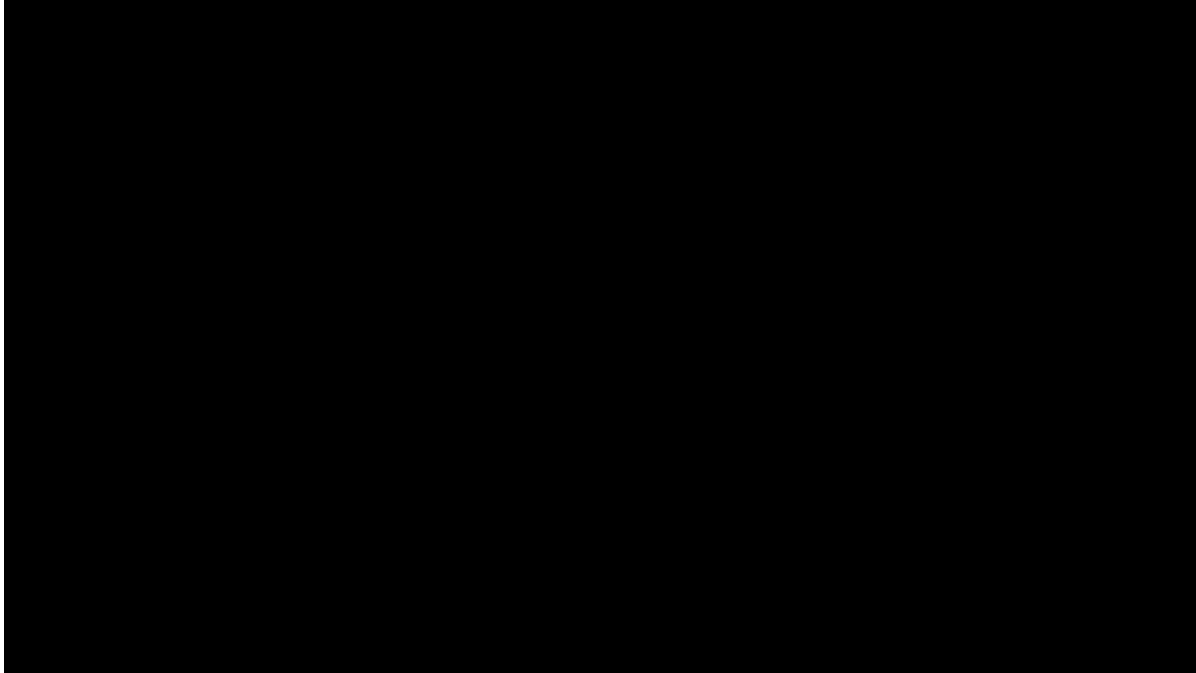


What are Microplastics?

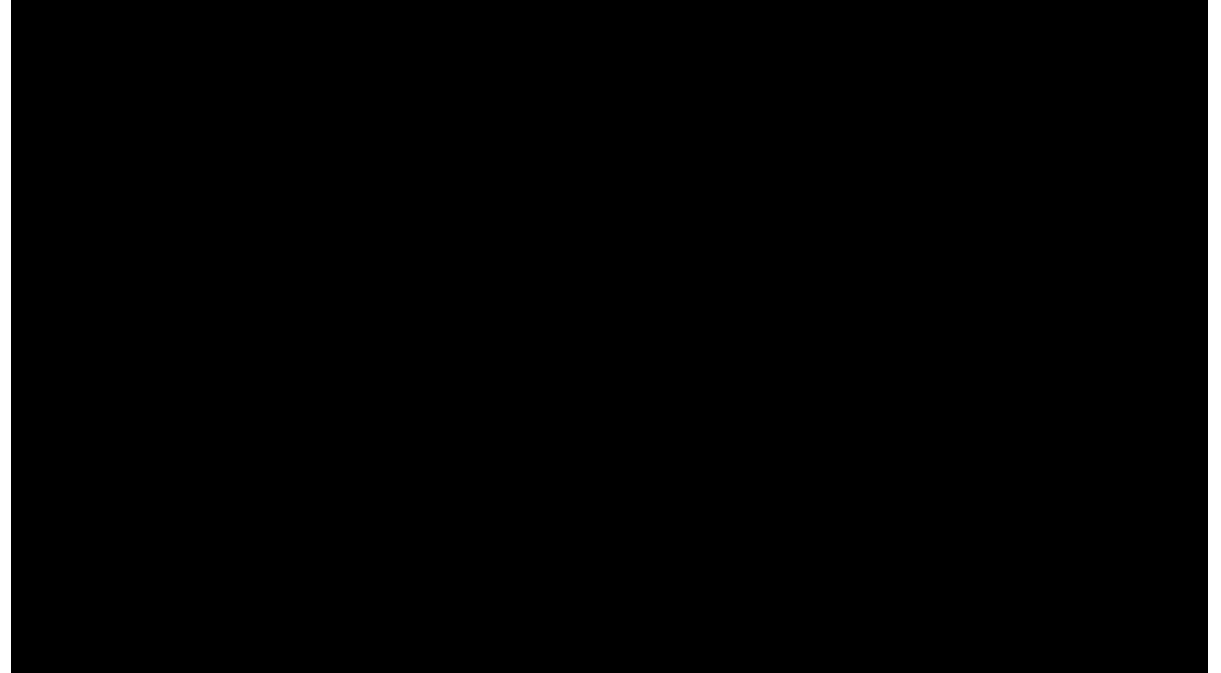


https://www.youtube.com/watch?v=aiEBEGKQp_I

Microplastics – an engineering challenge



<https://youtu.be/xHQrSOwRrow?si=dDzpUQN2RA6uE933>



<https://youtu.be/Y9MEyvsF0so?si=Ymgw5ZhOEHQQFyty>

Module 3

- Module 3 lab document is posted to Canvas
- Your team's task and video (or not) assignment is on the first page
- Now is a good time to meet with your group to find a time to sign up!